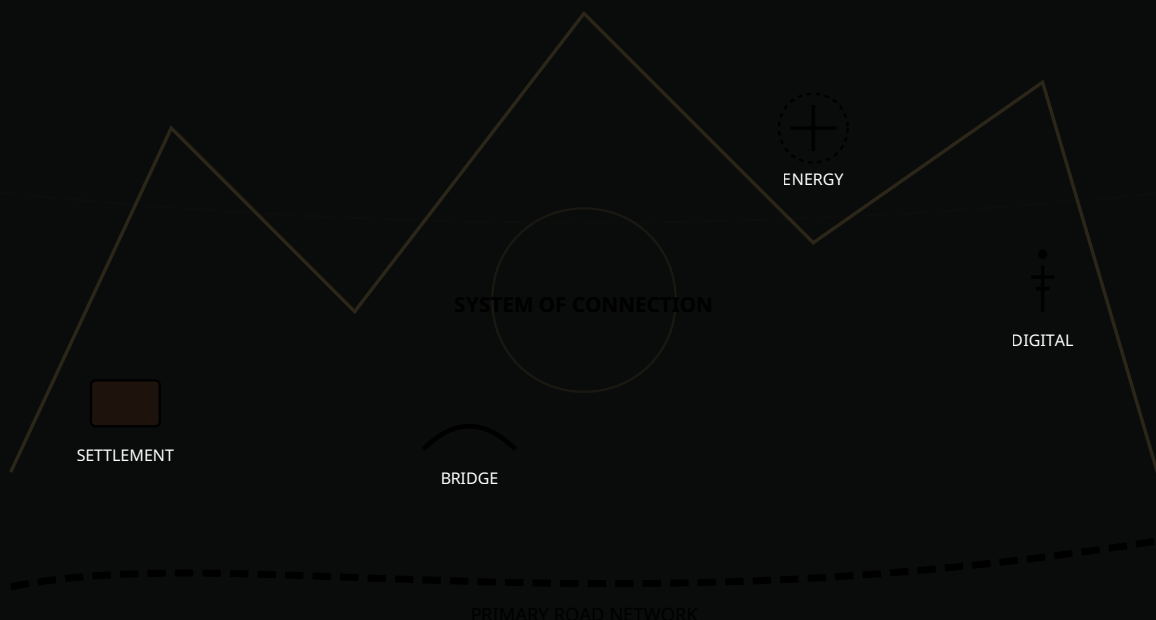


INFRASTRUCTURE & CONNECTIVITY

“Building the connections between people, places and opportunity.”

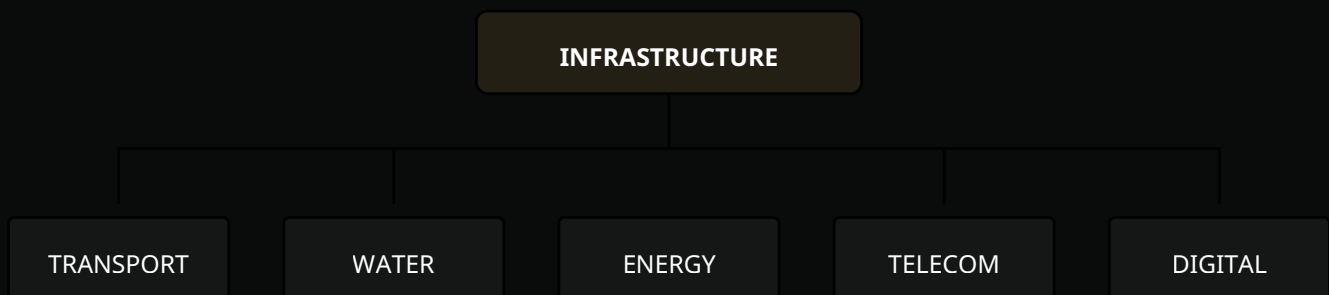
An infrastructure map can tell a story that a political map cannot. A road reveals where people travel. A bridge reveals where geography once created a barrier. An electricity line reveals where modern life has reached. A water system reveals how communities manage one of the most basic requirements of life. And a mobile signal can connect a mountain settlement to a person thousands of kilometres away. For Orakzai, infrastructure has always been connected to geography. Mountains create distance. Distance creates cost. Cost affects access. And access shapes opportunity. The story of infrastructure is therefore also a story of connection—connecting villages to towns, students to schools, patients to healthcare, farmers to markets, families to one another, and Orakzai to the wider world.



“The multi-layered infrastructure of the Orakzai highland.”

WHAT IS INFRASTRUCTURE?

Infrastructure includes the physical and digital systems that support society. It is not just about construction; it is the framework that allows people to move, goods to be traded, and services to be delivered. In Orakzai, infrastructure must overcome the mountain barrier to connect isolated valleys with regional hubs.^{3 6}



“The sectoral pillars of Orakzai's infrastructure development.”

CONNECTIVITY AS A SYSTEM

Infrastructure systems depend on one another. A road without maintenance becomes unreliable; internet without electricity becomes difficult to use; and a market without transport has limited reach. True connectivity is achieved when these layers are integrated into a single functioning system.⁷

ROAD + ENERGY + WATER + TELECOM + DIGITAL **CONNECTED COMMUNITY**

“The logical summation of a connected community.”

GEOGRAPHY & INFRASTRUCTURE: THE ENGINEERING CHALLENGE

Orakzai's geography—characterized by steep slopes, high peaks, and seasonal watercourses—presents unique engineering challenges. Infrastructure development in this terrain requires higher investment for retaining walls, culverts, and bridges. Geography creates the barrier; engineering creates the bridge.^{4 6}

MOUNTAIN — ENGINEERING CHALLENGE —> COST —> ACCESS

“The causal chain from geography to community access.”

INFRASTRUCTURE DEVELOPMENT (2026 STATUS)

As of June 2026, the Khyber Pakhtunkhwa government has approved a significant development package for Orakzai, prioritizing roads and water resources. These are programmed projects representing the state's commitment to post-merger integration.^{1 3}

PROJECT / SECTOR	STATUS (2026)	ALLOCATION (RS)
Road Infrastructure	ALLOCATED	9.45 Billion
Khan Mir Khel Dam	DESIGN/CONST.	7.40 Billion
Rural Access Roads	REHABILITATION	WB Package-1
Rescue 1122 Station	APPROVED	Tehsil Ismailzai

ROAD MAINTENANCE VS. BUILDING

A critical distinction in Orakzai’s development is the difference between building new infrastructure and maintaining existing assets. Mountain roads are subject to erosion, landslides, and seasonal damage, requiring consistent maintenance budgets to ensure they remain all-weather routes.⁴



“The path from construction to long-term infrastructure resilience.”

AUTHOR'S REFLECTION

The story of infrastructure in Orakzai is a story of connection. When I look at an infrastructure map, I don’t just see lines; I see the lived reality of our community. A road is not just asphalt; it is the student reaching school, the patient reaching the clinic, and the farmer reaching the market. The transition from a landlocked frontier to a connected district is the great task of our

generation. We must ensure that the connections we build today—whether they are roads of stone or networks of light—are strong enough to carry the potential of every Orakzai child into the wider world. Connection is dignity.

“CONNECTION IS DIGNITY. INFRASTRUCTURE IS THE PATH.”

INFRASTRUCTURE & CONNECTIVITY ANALYSIS